Optical matching and beam loss in the Korea Light Source Booster-to-Storage transfer line

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Abstract

A booster-to-storage-ring transfer line must deliver the extracted booster beam while satisfying the optical requirements of storage-ring injection. This study optimizes the Korea Light Source Booster-to-Storage line, using its baseline quadrupole settings as the reference. The baseline setting does not satisfy the prescribed exit lattice functions and has insufficient nominal clearance at the injection-septum blade. Optical matching and vertical-beta optimization restore the required exit conditions and improve the beam envelope and aperture margin. The remaining design problem is then examined among matched settings, for which fixed boundary optics preserve the nominal endpoint beam sizes. Static trajectory correction reduces loss, after which booster shot jitter governs the remaining blade interception. A Gaussian loss model validated against particle tracking identifies how the horizontal phase advance redistributes this jitter between exit position and angle. A setting selected directly for loss reduction improves transmission on independent error realizations, while a dispersion-free exit further reduces predicted blade loss at the expense of increased angular jitter. The results distinguish the initial improvement obtained by matching the baseline BTS line from the subsequent optimization of loss sensitivity. The conclusions are conditional on the modeled aperture geometry and error budget; the dispersion-free result is analytical, and storage-ring capture is outside the scope of the study.

Keywords: beam dynamics; error analysis; Korea Light Source; transfer line

1 Introduction

Diffraction-limited storage rings require control of the beam distribution delivered to the injection region [1]. The booster-to-storage-ring (BTS) transfer line connects the extracted booster beam to the prescribed injection conditions of the storage ring. Its optical design must accommodate the incoming emittance and momentum spread, satisfy the exit lattice functions, and maintain adequate clearance through the magnets and injection septum. At the septum, the beam envelope and trajectory must fit within a channel bounded by the blade; downstream, both position and angle affect the conditions delivered to the injection kicker. The 4 GeV Korea Light Source (KLS), described in its technical design report (TDR) [2] and in Refs. [3, 4], provides the application considered here. In the TDR design, the BTS line consists of two dipole and eight quadrupole magnets, and the storage ring is filled by off-axis injection with a four-kicker orbit bump and an in-vacuum thin septum with a 1 mm blade [2]. The line studied here is a different design version of the BTS, with three main dipoles and nine quadrupoles (Section 2), and its injection geometry is taken from our study of nonlinear-kicker magnet (NKM) injection into the KLS storage ring [5]; the results apply to this version of the line.

Previous transfer line studies establish the importance of treating the accelerator interfaces and available geometry together. The HEPS high-energy transport-line design matches the

injection-point lattice functions while accommodating the relative positions of the booster and storage ring, equipment clearances, and the selected injection scheme [6]. A subsequent HEPS study updates the transfer lines in response to the evolving accelerator layouts and optics [7]. For Diamond-II, the transfer lines are designed to preserve bunch properties with low loss while retaining diagnostic capability and flexibility for matching; the BTS modifications connect the new booster extraction geometry to the storage-ring septum [8]. The ESRF–EBS rematching similarly includes injection-region fields and mechanical constraints [9]. These examples show why an optical target and the physical restrictions along the transport path must be specified together.

Beam characterization and trajectory control address different parts of this matching problem. The SLS 2.0 transfer line includes a nondispersive section for quadrupole-scan measurements and a corrector configuration for separate control of injection position and angle [10]. Its commissioning results connect beam characterization to downstream matching and trajectory stabilization. The Sirius injection study reports efficiencies separately for the injector stages and examines booster realignment, emittance exchange, and remaining reproducibility issues [11]. Separating these stages prevents a transfer-line improvement from being equated with the same change in storage-ring injection efficiency. It also motivates evaluating a nominally matched line under incoming beam fluctuations and residual trajectory errors, rather than relying on envelope plots alone.

The requirements imposed by the injection device provide an additional design constraint. Pulsed-sextupole injection at MAX IV relates the incoming beam distribution to the nonlinear injection process [12]. A recent TPS study proposes a new booster-to-storage-ring line for nonlinear-kicker injection and identifies existing-ring geometry, trajectory correction, and injected-beam characterization as design considerations [13]. This is a closely related transfer-line application; the link between transfer-line delivery and downstream nonlinear acceptance is therefore directly relevant to the present work. Together, these studies motivate distinguishing nominal optical matching, clearance at the limiting restriction, and sensitivity to the position and angle delivered to the injection system. The present study examines those distinctions within a fixed BTS model with explicit aperture and jitter models.

A common transfer-line design objective is to reduce the peak beta functions and thereby increase aperture margin. For fixed emittances, momentum spread, and matched entrance and exit lattice functions, however, quadrupole rematching preserves the nominal beam sizes at the boundaries. Restrictions in the fixed sections upstream and downstream of the quadrupoles can therefore limit the line even when its interior envelopes are reduced. The same boundary conditions leave freedom in the betatron phase advance, which changes the transport of booster position and angle jitter. A matched setting can consequently change the loss probability at a septum blade without changing its nominal clearance. Distinguishing these mechanisms is necessary to determine whether a proposed optics improvement addresses the limiting restriction or only redistributes margin elsewhere in the line.

The goal of this study is to optimize the baseline KLS BTS setting to satisfy the prescribed exit lattice functions and improve beam clearance, and then to reduce loss sensitivity within the matched solution set. The baseline setting is the nine-quadrupole configuration before matching. Bounded optical matching followed by vertical-beta optimization produces the optimized setting shown in the nominal comparisons. A matched intermediate setting, referred to below as the matched reference, is used to isolate the effect of reducing beta while preserving the same exit target. The line is modeled with element-wise apertures and the injection-septum blade geometry of the NKM injection study [5]. The subsequent error study compares the matched reference, low-beta, and loss-based settings after static trajectory correction using field trims of existing dipoles. A Gaussian loss probability derived from the transported beam and centroid covariance is validated against particle tracking and used to select a loss-based setting before evaluation on independently seeded error realizations. The analysis also examines the reachable phase advance,

a dispersion-free exit target, and sensitivity to the error levels. The study ends at the transfer-line exit; storage-ring capture and NKM transparency require the subsequent injection calculation.

The paper is organized as follows. Section 2 defines the BTS lattice, beam covariance, apertures, and trajectory-correction model. Section 3 presents the clearance and loss criteria, matched-end jitter transport, and numerical validation procedures. Section 4 compares the aperture restrictions, corrected trajectories, loss-based setting, exit targets, and error sensitivities. Section 5 discusses the design implications and limitations.

2 Parametric model and physical observables

2.1 Lattice, coordinates, and boundary conditions

The baseline BTS model contains 36 elements over 21.789 m, from marker m1 at the entrance to marker m4 downstream of the injection septum. Beam dynamics simulations use the Accelerator Toolbox (AT) [14] through its Python implementation pyAT [15]. The line has nine 0.2 m quadrupoles in three triplets, three main dipole bending magnets (`b1`, `b2`, and `b3`, in beam order), the booster extraction kicker `kext`, the extraction septum `sept_in` and the injection septum `sept_ex`; the kicker and both septa are modeled as dipole elements (Fig. 1). Quadrupoles are labeled `qij`, where i identifies the triplet and j the magnet within that triplet, both in beam order. Thus `q11` is the first quadrupole in the first triplet and `q33` is the last quadrupole in the third triplet. Local coordinates are $(x, p_x, y, p_y, \delta, ct)$ with positions/path length in m, normalized canonical momenta approximately angles in rad, and $\delta = \Delta p/p_0$ dimensionless. The six design bend angles are 7.5 mrad, 88.5 mrad, −111.701 mrad, 176 mrad, −111.701 mrad and 88.5 mrad; reference geometry is held fixed when fields or particle momenta change.

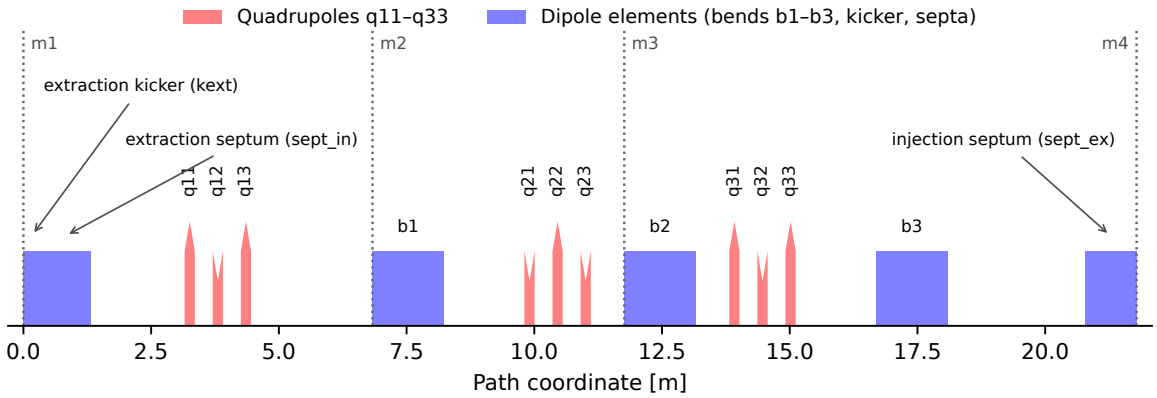


Figure 1: Baseline BTS element sequence drawn with Accelerator Toolbox synoptic plotting. Red symbols are the quadrupoles `q11`–`q33`; blue blocks are the dipole elements: the extraction kicker `kext`, the extraction septum `sept_in`, the main bends `b1`–`b3` and the injection septum `sept_ex`. Blank intervals are drifts. Dotted lines mark the markers m1–m4 used for boundary and intermediate optics observations.

Entrance and exit lattice functions (Table 1) are inherited design values. The exit m4 is the injection-septum exit: the NKM injection study uses exactly these values as its injected beam and transports it 2.26 m to the kicker. The configured nonzero exit dispersion (80.9 mm, 47.5 mrad) differs from the ring dispersion at the injection kicker, which is about 0.3 mm; we therefore also examine a dispersion-free exit. Quadrupole family bounds, bore radius, pole-field limit, the 60 m beta screening ceiling and square aperture limits are model parameters. The BTS line is independently validated using analytical maps of the same AT element sequence.

Table 1: Prescribed entrance and exit lattice functions. The exit values describe the injected beam at the injection-septum exit used in the NKM injection study [5]. A dispersion-free exit is examined in Section 4.4.

Observable	m1 entrance	m4 target
β_x, β_y [m]	7.560000, 12.269000	2.336495, 4.256241
α_x, α_y [1]	1.523100, −1.654700	−0.016335, 0.017772
D_x [m], D'_x [1]	0.276200, −0.065700	0.080868, 0.047472

The four-station comparison model places square apertures at markers m1–m4 (19.35 mm at the line ends, 30 mm at two interior markers) and none on the magnets. That four-station model provides a comparison with the element-wise restrictions of Section 2.3.

2.2 Beam covariance and clearance

This study uses the booster horizontal design input $\epsilon_x = 7.67\,\text{nm rad}$ and $\sigma_\delta = 0.106\,\%$ [4]. The vertical input $\epsilon_y = 0.767\,\text{nm rad}$ corresponds to a vertical-to-horizontal emittance ratio of 10 %. Betatron coordinates and momentum are independent untruncated Gaussians before dispersion is added; initial transverse coupling is zero and $ct = 0$ for this RF-free single pass. The transverse blocks of the complete covariance are

$$\Sigma_{u,\text{bet}} = \epsilon_u \begin{pmatrix} \beta_u & -\alpha_u \\ -\alpha_u & (1+\alpha_u^2)/\beta_u \end{pmatrix}, \qquad \Sigma_0 = \Sigma_{\text{bet}} + \sigma_\delta^2 \mathbf{d}\mathbf{d}^\mathsf{T}, \tag{1}$$

where $u \in \{x, y\}$ denotes the transverse plane, β_u and α_u are the Courant–Snyder parameters and ϵ_u the emittance in that plane, Σ_{bet} is the 6×6 betatron covariance with the two 2×2 blocks $\Sigma_{u,\text{bet}}$ on its diagonal, and $\mathbf{d} = (D_x, D'_x, D_y, D'_y, 1, 0)^\mathsf{T}$. The same index u is used for the plane in all equations below. Rolls of the quadrupoles about the beam axis (Section 2.4) couple the horizontal and vertical planes downstream of each rolled magnet. Full covariance transport, $\Sigma_s = M_s \Sigma_0 M_s^\mathsf{T}$, retains that coupling and momentum correlation. A local affine map gives the centroid $\bar{\mathbf{z}}_s = M_s \bar{\mathbf{z}}_0 + \mathbf{b}_s$.

For each aperture station s and plane u, with lower and upper aperture edges a_- and a_+, we define

$$c_u(s) = \min[a_+ - \bar{u}(s), \bar{u}(s) - a_-] - 5\sqrt{\Sigma_{uu}(s)}, \qquad c_{\min} = \min_{s \in \text{stations}, u \in \{x,y\}} c_u(s). \tag{2}$$

Here $\bar{u}(s)$ is the beam centroid and $\sqrt{\Sigma_{uu}(s)}$ the projected rms beam size, including the dispersive contribution. The first term in Eq. (2) is the distance from the centroid to the nearer aperture edge; subtracting five rms beam sizes gives the margin $c_u(s)$ that remains between the five-sigma beam envelope and that edge. A positive c_u means that the envelope fits inside the aperture, $c_u = 0$ that it touches the edge, and a negative c_u that it overlaps the edge by $|c_u|$. The minimum $c_{\min}$ over all stations and both planes identifies the limiting restriction. The five-sigma envelope is a conventional coverage measure of the projected distribution; an untruncated Gaussian beam is never completely contained. Because the beam size scales as $\sqrt{\beta_u}$, a reduction of the beta function by some metres does not increase the clearance by a comparable length: the clearance changes only by five times the change of the rms size at the limiting station, and not at all where the optics are fixed by the boundary conditions (Section 3). Historical optics-only ensembles use a separate conservative 100 nm rad/10 nm rad, 0.11 % beam configuration and are not pooled with this study.

The quadrupole strengths are bounded so that every setting considered remains within the field range of conventional quadrupole magnets. At 4 GeV the magnetic rigidity is $B\rho = p_0/e \approx 13.342\,56\,\text{T m}$. A normalized strength K in m^{-2} corresponds to the field gradient $G = K(B\rho)$ in

$\mathrm{T\,m^{-1}}$, and a quadrupole with bore radius r then has the pole-tip field $B_{\mathrm{pole}} = |G|r$ in T. The strength bounds are $|K| \leq 2.5\,\mathrm{m^{-2}}$ for the outer quadrupoles q11 and q33 and $|K| \leq 3\,\mathrm{m^{-2}}$ for the other seven. With $r = 19.35\,\mathrm{mm}$, the largest gradients are $33.356\,\mathrm{T\,m^{-1}}$ and $40.028\,\mathrm{T\,m^{-1}}$, and the corresponding pole-tip fields are 0.645 T and 0.775 T, respectively. Both remain below the 1.2 T pole-tip limit used for screening. This check confirms that the optimized strengths are consistent with the magnet model; it is not an engineering design of the magnets.

2.3 Element-wise restrictions

Every element of the BTS line carries its own aperture (Table 2), and AT checks each aperture at the element entrance. Interior envelopes are sampled for the nominal profile on an observation-only copy of the line, with four subdivisions per quadrupole and eight per bend or septum.

The injection-septum blade separates the main-field channel traversed by the injected beam from the stored-beam region, where a small magnetic field is required. Its geometry follows the NKM injection study [5]; for comparison, the baseline KLS injection uses an in-vacuum thin septum with a 1 mm blade [2]. Let X denote the horizontal coordinate relative to the storage-ring reference orbit at the septum exit. The blade face toward the stored beam is at $X = -16\,\mathrm{mm}$ and its thickness is 2 mm, so the face toward the injected beam is at $X_b = -18\,\mathrm{mm}$. The nominal injected-beam centroid is at $X_{\mathrm{inj}} = -19.478\,\mathrm{mm}$, giving the centroid-to-blade distance

$$d_b = X_b - X_{\mathrm{inj}} = 1.478\,\mathrm{mm}. \tag{3}$$

For the configured beam, this distance is approximately three rms horizontal beam sizes plus 1 mm. In the local BTS coordinate at the exit, $x = X - X_{\mathrm{inj}}$, the nominal injected beam is centered at $x = 0$ and the blade face is at $x = +d_b$. This boundary is applied on the side facing the stored beam in the injection septum and at the line end (marker m4). The five-sigma clearance in Eq. 2 additionally subtracts the projected beam envelope and includes any centroid displacement; it is distinct from d_b.

The injected beam is intentionally deflected by the septum's main magnetic field while it travels through the injection channel. Proximity to the blade does not imply that the beam is in a field-free region. In the BTS model, the septum's bending field is represented by a dipole element, while the blade face is represented separately by the aperture boundary. Detailed spatial fringe-field and leakage-field distributions are not resolved. The blade dimensions and offset are study parameters rather than established hardware specifications; the other aperture dimensions are model parameters, and the bend gap and the extraction-septum aperture are swept (Section 4).

Bend chambers are taken as curved: the sagitta of the reference arc over its chord is 30.8 mm in b2 and 19.5 mm in b1/b3, so a straight chamber would need that much extra width. Clearances (Eq. 2) are evaluated at every aperture, with the projected halfwidth of a round aperture reduced by the other plane's centroid.

Table 2: Element-wise aperture model, reported as halfwidths. For the injection septum, a_x is the nominal injected-centroid-to-blade distance on the $+x$ side, not the blade thickness or its offset from the storage-ring reference orbit. The outer wall is at $-8.5\,\mathrm{mm}$ in local BTS coordinates.

Region	Shape	a_x [mm]	a_y [mm]
Drifts	round	19.350	19.35
Quadrupoles q11–q33	round	19.350	19.35
Extraction kicker	rect.	15.000	10.00
Extraction septum	rect.	7.500	5.00
Bends b1–b3 (curved)	rect.	30.000	10.00
Injection septum	one-sided	1.478	5.00

2.4 Errors, jitter and trajectory correction

Static machine errors are independent Gaussian errors of the nine quadrupoles, namely gradient errors (0.1 % rms), x/y offsets (100 µm rms) and rolls about the beam axis (0.5 mrad rms), together with field errors of the six dipole elements (0.05 % rms). Each independent draw of these static errors defines one error realization of the line. Each shot adds booster jitter at m1: entrance beta factors (5 %), centroid position 0.5 mm, angle 0.2 mrad and momentum 0.1 %, the latter carried by the entrance dispersion. These levels are model inputs; the booster jitter is the same as in the NKM injection study [5]. Their sensitivity is examined in Section 4.5.

Static trajectory errors are corrected without changing the line: no corrector or monitor element is added. The horizontal knobs are field trims of the six existing dipoles, applied as additional fixed-geometry dipole fields like the dipole errors. Horizontal readings are taken at existing element boundaries, after each quadrupole, at the injection-septum entrance and at the line end, with a static 50 µm rms reading error (offset and calibration residual). The trims minimize the weighted readings (septum entrance and end weighted 3) with the response of the error-free model line. With $\mathbf{r}$ the vector of readings, A the response matrix of the readings to the trims $\boldsymbol{\theta}$, and W the diagonal weight matrix, the trims minimize

$$\|W(\mathbf{r} + A\boldsymbol{\theta})\|^2 + \lambda^2\|\boldsymbol{\theta}\|^2. \tag{4}$$

The second term is a Tikhonov (ridge) regularization [16]: it penalizes large trims and keeps the solution stable when the response matrix is ill-conditioned. Here λ is 10^{-3} times the largest singular value of WA. The trims are limited to ± 1 mrad of kick, and the correction is applied in two passes. The correction uses no knowledge of the actual errors and does not act on shot jitter. The line has no vertical knob, so the vertical plane is not corrected. Every error realization is evaluated uncorrected and corrected on identical particles.

3 Physical criterion, validation and comparison protocol

3.1 Active restrictions and boundary-fixed clearances

The two end stations sit where the optics are fixed by boundary data: the entrance Courant–Snyder parameters and dispersion are inputs, and every admissible setting matches the exit to its target. For a centered nominal beam the clearance there is therefore $c_u = a - 5\sqrt{\epsilon_u\beta_u + D_u^2\sigma_\delta^2}$ with the boundary values of Table 1, for every quadrupole setting K that satisfies the match. With $a = 19.35$ mm this gives 17.455 mm/18.865 mm at the entrance and 18.555 mm/19.064 mm at the exit (x/y), exactly the values found numerically for both settings in the four-station model. The two interior stations have 30 mm halfwidths and could limit only if a five-sigma size exceeded about 12.5 mm, which no matched setting approaches. In this four-station model the nominal minimum clearance is thus analytically independent of K: the "saturation" of the minimum is a consequence of placing restrictions only where the optics are fixed, not a finding about the optics. Under errors, the only K-dependent part of the end-station clearances is the response of the centroid at the exit to errors and jitter. The same holds for any restriction upstream of q11 or downstream of q33: there the optics are fixed by the boundary data, so only the trajectory response can depend on K.

At a single symmetric active station away from $\bar{u} = 0$, the local sensitivity is

$$\frac{\partial c_u}{\partial K_j} = -\operatorname{sgn}(\bar{u})\frac{\partial \bar{u}}{\partial K_j} - \frac{5}{2\sigma_u}\frac{\partial \Sigma_{uu}}{\partial K_j}. \tag{5}$$

Switching restrictions or crossing zero centroid makes this expression piecewise. In an uncoupled plane $\Sigma_{uu} = \epsilon_u\beta_u + D_u^2\sigma_\delta^2$; our coupled calculation retains the complete covariance. Reducing beta controls one size contribution, not necessarily dispersion or orbit sensitivity. We export these variances and centroids separately, adding variances rather than RMS sizes.

3.2 Jitter transfer between matched ends

Because both ends are matched, the horizontal transfer matrix from m1 to m4 is

$$M_x = B_4 R(\mu_x) B_1^{-1}, \qquad B = \begin{pmatrix} \sqrt{\beta} & 0 \\ -\alpha/\sqrt{\beta} & 1/\sqrt{\beta} \end{pmatrix}, \tag{6}$$

with $R(\mu_x)$ a rotation by the horizontal phase advance and $B_{1,4}$ the boundary Courant–Snyder parameters of Table 1. A shot centroid at m1 with covariance J (position 0.5 mm, angle 0.2 mrad) therefore reaches the exit with covariance $M_x J M_x^{\mathsf{T}}$ plus the momentum term $\sigma_{\delta,j}^2 \mathbf{d}_4 \mathbf{d}_4^{\mathsf{T}}$, which is the same for every matched setting. The quadrupoles influence the exit jitter only through μ_x. Over a full period the exit position spread ranges from 0.26 mm to 0.95 mm and the angle spread from 0.12 mrad to 0.41 mrad, in opposite phase (Fig. 4, left): a matched line can move booster jitter between position, which the septum blade sees, and angle, which the injection kicker sees, but cannot remove it. Steered static errors add a small residual (Section 4.2).

3.3 Loss-based observable and design

For a Gaussian projection with centroid $\bar{u}$ and rms size σ at an aperture with edges $a_\pm$, the lost fraction is

$$P = \Phi\left(\frac{a_- - \bar{u}}{\sigma}\right) + \Phi\left(\frac{\bar{u} - a_+}{\sigma}\right), \tag{7}$$

with Φ the standard normal distribution function. Per shot, the line loss is bracketed by the largest single-aperture P and the union bound; we report the former as the prediction. When the centroid is itself Gaussian, as for linearly transported jitter, the shot-averaged lost fraction is Eq. (7) with the summed covariance of beam and jitter. This makes a smooth, loss-relevant objective. The loss-based setting minimizes the logarithm of the largest shot-averaged P on the error-free line, subject to the six exit equalities, horizontal peak beta ≤ 60 m and the strength bounds, using sequential least-squares programming (SLSQP) [17] with a 200-iteration budget and termination tolerance 10^{-10}. Three initial settings are evaluated in the bounded search. The setting is frozen before a fresh test on 200 steered error realizations, evaluated with tracked particles.

3.4 Accuracy and spatial observation resolution

Exact independent hard-edge maps are checked against AT tracking at every element boundary. Tracking uses 40 integration steps per magnet, with an absolute nominal-map maximum-entry tolerance of 2×10^{-6} in the stated SI coordinate convention. Table 3 summarizes the accuracy checks. Transverse symplectic discrepancy is 1.22×10^{-15}, without a longitudinal six-dimensional certificate. Translations, rolls and fixed-geometry dipole-field perturbations are exercised independently. Momentum errors enter particle δ without double rescaling magnet fields.

The affine error maps of the earlier four-station campaign omit chromatic derivatives about displaced trajectories. Comparing all 600 shots of that campaign with AT Jacobians expanded about the actual centroid changes station clearances by up to 78.43 µm, but shifts the paired mean by only -0.155 µm. Eight matched-reference and three alternative active-station classifications change. The earlier paired mean of 14.61 µm and its interval $[-17.91, +48.32]$ µm become 14.45 µm and $[-17.88, +48.10]$ µm; the unresolved-benefit conclusion persists. New comparisons use centered AT maps. Moment probes are uncensored by aperture limits, so propagated moments describe the incoming distribution rather than a survivor-biased subset.

Representative simultaneous-error cases compare probe steps 10^{-6}–10^{-8} and forty/eighty integration steps. Their clearance changes are below 5×10^{-6} µm. Product Gauss–Hermite quadrature at orders three, five and seven propagates the unrestricted Gaussian through nonlinear AT; five/seven convergence is negligible on the checked cases. Centered-linear versus nonlinear-moment discrepancy is at most 0.606 µm on four extreme or representative shots of the earlier

campaign and 0.6858 µm on two new error realizations for every regime/design. These finite-case checks are smaller than the reported effect intervals and illustrative 0.1 mm decision scale; they are not universal error bounds or Gaussian loss-tail certificates. Intentional scraping verifies first-loss accounting.

Spatial observation is refined independently of integration steps. Exact maps sample each finite element at 8, 16, 32 and 64 subdivisions, applying bend edges only once. Matched-reference boundary peaks 31.44 m/50.75 m become 31.63 m/51.27 m; old low-beta peaks 33.58 m/28.18 m become 33.77 m/28.50 m. Reoptimizing vertical peak beta on sixteen subdivisions converges in 34 SLSQP iterations. The resulting 64-subdivision peaks are 33.79 m/28.44 m; changes from 32 subdivisions are below 0.35 mm. Table 4 reports the refined setting at each observation resolution. These are spatially converged sampled maxima, not exact analytical extrema or a global optimum. The second quadrupole in the third triplet, `q32`, remains at its lower bound $-3\,\mathrm{m}^{-2}$.

Table 3: Numerical accuracy checks for the nominal transport and selected error cases. Map-entry discrepancies use the stated SI coordinate convention. The final four rows are finite-case convergence diagnostics, not universal error bounds or Gaussian loss-tail tolerances.

Check	Discrepancy	Tolerance / scope
Nominal map-entry discrepancy	1.58×10^{-8}	2×10^{-6}
Transverse symplectic discrepancy	1.22×10^{-15}	1×10^{-10}
Probe-step clearance change	1.85×10^{-6} µm	Finite-case diagnostic
40/80-step clearance change	4.12×10^{-6} µm	Finite-case diagnostic
Centered-map/nonlinear clearance difference	0.606 µm	Finite-case diagnostic
Order-5/7 quadrature clearance change	3.32×10^{-11} µm	Finite-case diagnostic

Table 4: Spatial convergence of the refined low-beta setting. Integration remains at 40 steps per magnet while the observation mesh changes. The reported peaks are sampled maxima; the last column is the larger change in either plane relative to the preceding mesh.

Subdivisions	Peak β_x [m]	Peak β_y [m]	Max. change [m]
8	33.791374	28.436689	—
16	33.791374	28.436689	9.97×10^{-9}
32	33.794522	28.436713	0.003 15
64	33.794522	28.437054	0.000 341

4 Results

4.1 Active restrictions of the element-wise line

Figures 2 and 3 compare the baseline BTS setting with the optimized low-beta setting. The reference uses the baseline quadrupole strengths from the BTS configuration, before optical matching. Its exit lattice functions miss the prescribed target, and its nominal five-sigma envelope extends across the injection-septum blade boundary. The optimized setting satisfies the exit target, reduces the large vertical envelope, and restores positive nominal clearance. Table 5 gives the corresponding exit lattice functions, peak beta values and minimum clearances. The figure comparison therefore includes the improvement from matching the baseline BTS line, rather than holding the exit optics fixed for both curves.

After matching, a separate comparison of the matched reference and optimized low-beta setting isolates the effect of interior beta reduction. In that matched comparison the injection-septum

Table 5: Nominal optimization of the baseline BTS setting. Peak beta functions are sampled with the same spatial resolution as Figure 2. The maximum normalized exit residual is evaluated using the declared optical scales; a matched setting requires it to be below 10^{-3}. A negative clearance means that the projected five-sigma envelope overlaps an aperture boundary, not that every particle is lost.

Observable	Baseline setting (reference)	Optimized low beta
Exit β_x [m]	44.9808	2.3365
Exit β_y [m]	242.6069	4.2562
Peak β_x [m]	52.2506	33.7945
Peak β_y [m]	242.6069	28.4371
Minimum clearance [mm]	-1.5170	0.6484
Max. normalized exit residual	4.77×10^{3}	2.88×10^{-10}

clearances are 0.648 mm at the entrance and 0.683 mm at the exit, essentially equal for both settings. The nearest interior restriction is the vertical gap of b2, with 9.12 mm for the matched reference and 9.32 mm for the optimized low-beta setting. The bend-gap and extraction-septum aperture sweeps leave the blade as the limiting restriction for these matched settings. Thus optimization of the baseline BTS line improves nominal delivery, while further reductions of interior beta at a fixed exit target act mainly on clearance away from the blade. The error-study tables below retain this matched reference as a separate comparison setting; they do not report an error ensemble for the baseline setting.

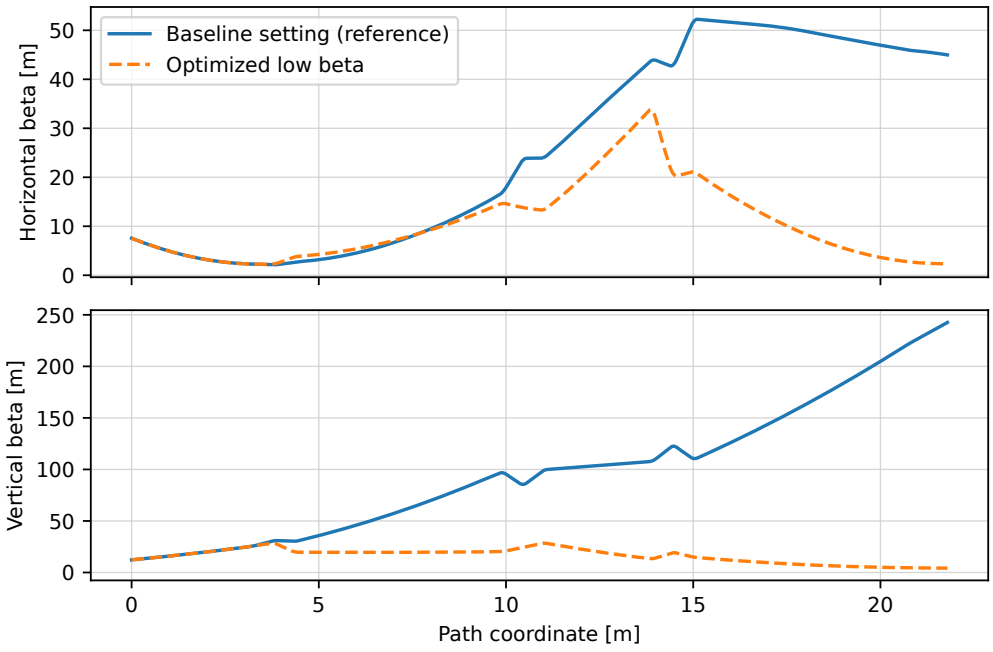


Figure 2: Baseline BTS setting (reference) and optimized low-beta lattice functions, drawn with Accelerator Toolbox linear-optics plotting from the same entrance Courant–Snyder parameters and with the same observation resolution. The baseline setting uses the unmodified configuration strengths and does not satisfy the exit target; the optimized setting is matched. Both curves use the same magnet positions and bend geometry.

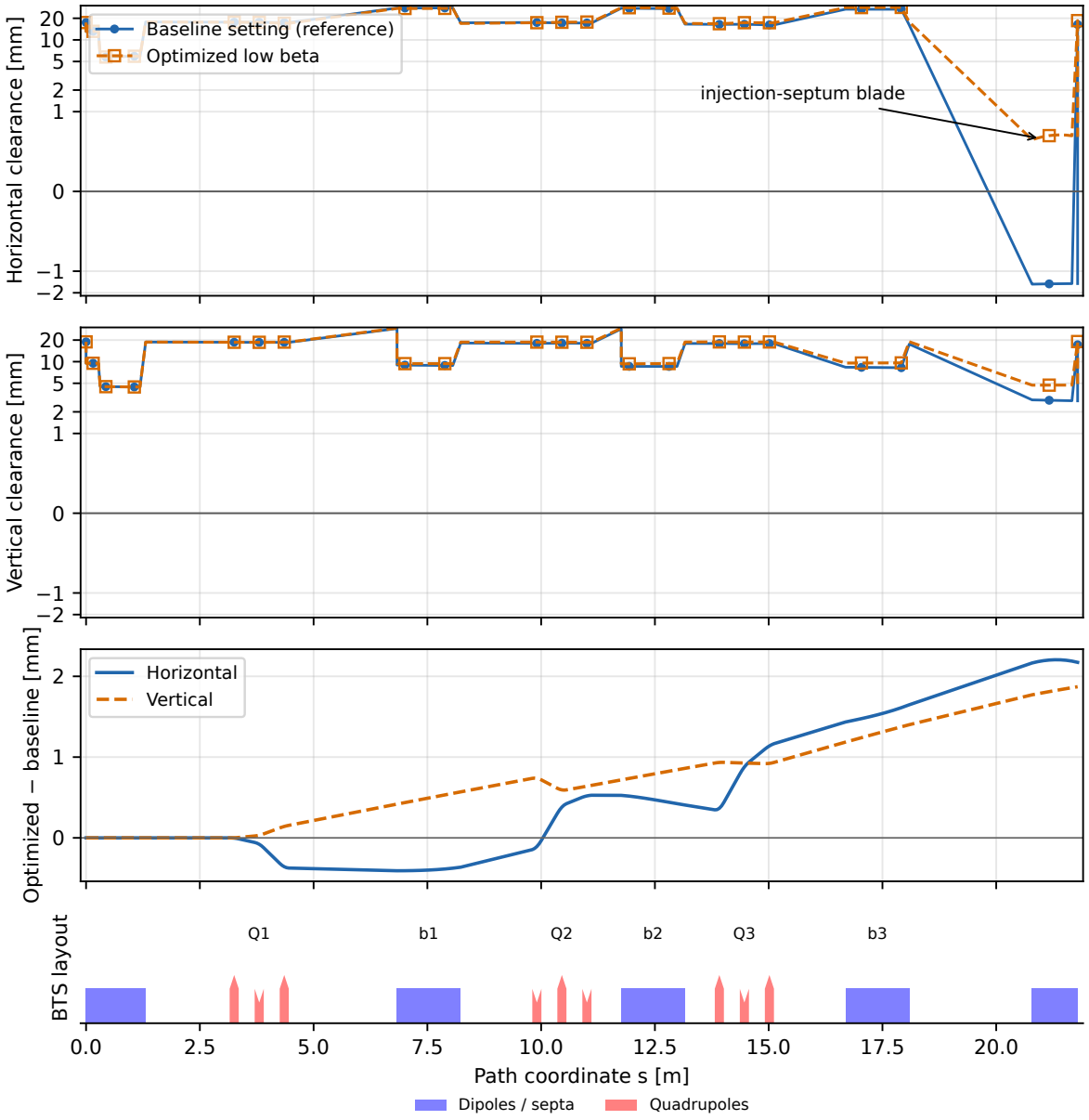


Figure 3: Nominal five-sigma clearance for the baseline BTS setting (reference, solid blue with filled circles) and optimized low beta (dashed orange with open squares). A symmetric logarithmic scale, linear near zero, displays the baseline setting's negative horizontal clearance at the injection septum. Zero marks the five-sigma envelope touching an aperture. The third panel shows optimized minus baseline clearance in each plane. The bottom panel gives the common BTS layout; Q1–Q3 identify quadrupole triplets. The baseline and optimized settings share the entrance conditions but have different exit lattice functions. Design beam, no errors.

4.2 Trajectory correction

Table 6 compares the uncorrected and corrected matched lines on 200 error realizations with three shots of 2 000 particles each; both settings and both variants see the same error realizations, shots and particles. Without correction the static errors displace the exit centroid by 1.2 mm rms, and 10.08 % of the particles are lost, almost all on the blade. With the dipole trims the exit position spread falls to 684 µm and the angle spread to 367 µrad, and the loss to 1.68 % (1.47 % for the low-beta setting); the rms trim is 204 µrad. Two trims close to the exit, b3 and the injection septum, steer both position and angle at the septum. The uncorrected vertical orbit stays far inside the vertical restrictions.

Table 6: Steering of the matched reference (Matched) and optimized low beta on 200 error realizations (3 shots, 2 000 particles each). $\bar{c}_{\min}$ is the mean minimum five-sigma clearance, Lost the tracked particle-loss fraction, Predicted the mean Gaussian loss probability of the most loss-prone aperture, Blocks the number of error realizations with any observed loss, and x_{rms}, x'_{rms} the rms exit centroid over shots.

Setting	Steering	$\bar{c}_{\min}$ [mm]	Lost [%]	Predicted [%]	Blocks	x_{rms} [mm]	x'_{rms} [mrad]
Matched	none	0.584	10.08	10.04	74	1.20	0.36
Low beta	none	0.556	10.00	10.01	79	1.20	0.38
Matched	dipole trims	0.524	1.68	1.68	47	0.68	0.37
Low beta	dipole trims	0.523	1.47	1.47	44	0.68	0.38

Two observations arise from these results. First, after correction the exit centroid spread, about 0.68 mm and 0.37 mrad, is mainly that of the booster jitter, and it sets the remaining blade loss. Second, the mean minimum clearance does not rank the variants by loss: the uncorrected line has the largest mean clearance (0.584 mm against 0.524 mm) and six times the loss, because a static offset away from the one-sided blade raises the mean while offsets towards it are lost. The Gaussian loss prediction, by contrast, agrees with the tracked loss within 0.4 % (relative) in every row. The matched reference and low-beta settings are indistinguishable in every column.

4.3 Reachable phase advance and the loss-based setting

Bounded least-squares matching from 360 random starts finds 349 matched quadrupole settings within the strength bounds. Their horizontal phase advance spans only 200.2°–232.0°: the exit match, with its dispersion targets and the bend layout, pins μ_x to an interval in which booster jitter falls largely on the exit position. The matched reference (211.8°) and low-beta (210.8°) settings lie near its middle. Within the interval the predicted shot-averaged blade loss varies by more than a factor of ten (Fig. 4, right) and decreases monotonically towards the lower edge.

The loss-based search ends at that edge, $\mu_x = 200.2°$. All three SLSQP starts reach the 200-iteration limit; only the start from the best pool member is feasible, and it improves that member by 1.0 %. The setting is therefore a feasible point at the boundary of the reachable interval, not a certified optimum. Its predicted shot-averaged loss is 0.27 % against 1.28 % for the matched reference, because its exit position jitter is 506 µm instead of 643 µm (Table 7).

On the fresh error realizations the tracked loss is 2.30 %, 2.27 % and 0.69 % for the matched reference, low-beta and loss-based settings, each within 1 % of its prediction. The loss-based setting changes the loss by -1.61 percentage points against the matched reference (pointwise 95 % interval clustered by error realization, [-2.33, -0.96] percentage points), a factor of 3.3; low beta changes it by -0.03 percentage points ([-0.23, +0.17] percentage points), unresolved. Its mean minimum clearance changes by only −0.017 mm ([-0.035, +0.001] mm), also unresolved: the mean-clearance metric cannot resolve this loss reduction. The price is an exit angle jitter larger by 13 % (368 µrad against 327 µrad), which the injection kicker must accept. The loss is concentrated in shots whose entrance jitter is two to three standard deviations: the five worst

Table 7: Matched reference (Matched), optimized low-beta and loss-based settings and their fresh test (200 steered error realizations, 3 shots, 2 000 particles each). μ_x: horizontal phase advance; $\hat{\beta}$: larger peak beta; $\sigma_{x,j}$, $\sigma_{x',j}$: exit jitter from Eq. (6); $\bar{P}$: shot-averaged loss on the error-free line; Lost and Pred.: tracked and predicted loss on the test realizations; Blocks: error realizations with any loss.

Setting	μ_x [deg]	$\hat{\beta}$ [m]	$\sigma_{x,j}$ [mm]	$\sigma_{x',j}$ [mrad]	$\bar{P}$ [%]	Lost [%]	Pred. [%]	Blocks
Matched	211.8	51.3	0.643	0.327	1.28	2.30	2.31	57
Low beta	210.8	33.8	0.631	0.331	1.16	2.27	2.28	55
Loss-based	200.2	48.1	0.506	0.368	0.27	0.69	0.69	29

of the 200 test realizations of the matched reference contribute 38 % of its loss, the worst losing 43 % per shot on average. The intervals, clustered by error realization, account for this heavy tail, and the population mean differs between independently seeded populations (Section 4.5).

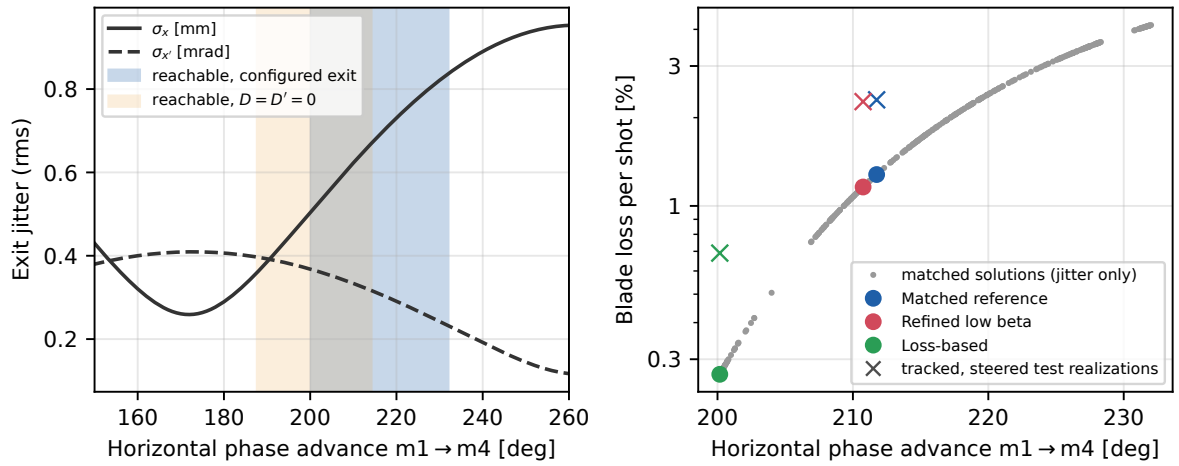


Figure 4: Left: rms exit position (solid, mm) and angle (dashed, mrad) jitter from Eq. (6) versus horizontal phase advance, with the reachable intervals for the configured exit (blue) and a dispersion-free exit (orange; its momentum term differs slightly). Right: predicted shot-averaged blade loss of every matched solution (grey) and of the three settings (circles), with the tracked loss on steered test realizations (crosses), which also include static residuals. The matched reference in this comparison is the intermediate matched setting; the baseline setting appears in Figures 2 and 3.

4.4 Exit target and dispersion

At m4 the configured target gives the injected beam an rms divergence of 76 µrad: 57 µrad from the betatron emittance and 50 µrad from $D'_x\sigma_\delta$. The 0.1 % shot energy jitter adds a centroid angle of 47 µrad and a position of 81 µm rms. Both terms are fixed by the target, the same for every matched setting, and absent for a dispersion-free exit.

Matching the same Courant–Snyder parameters with $D_x = D'_x = 0$ at m4 is feasible within the strength bounds: all 115 solutions from 120 bounded random starts keep both peak betas below 60 m (smallest larger peak 26.3 m). The dispersion-free target also widens the reachable phase advance to 187.7°–214.2° (Fig. 4), which includes phases where booster jitter falls mainly on the exit angle. The best of these solutions ($\mu_x = 187.7°$, peak betas 31.4 m/48.2 m) has a shot-averaged blade loss fraction of 4.1×10^{-5}, 65 times below the loss-based setting with the configured target. Its exit jitter is 350 µm and 394 µrad, and the beam divergence falls to

57 µrad. These numbers are analytic, from the same Gaussian model whose predictions agreed with tracking within 1 % above; the dispersion-free settings were not tracked on test realizations.

The exit target is thus the BTS design quantity with the largest leverage on injection. Removing the exit dispersion lowers the injected beam's divergence by a quarter and lets the line put booster jitter where the septum blade does not see it, but the angle jitter, 394 µrad against 327 µrad for the matched reference, then arrives at the injection kicker. In the NKM injection study [5], the kicker's 80 % angular acceptance is 0.44 mrad, and once the static vertical injection error left by transfer-line magnet errors is removed, booster position and angle jitter are the dominant kicker-specific capture losses. This line has no vertical correction; correcting that vertical error will be addressed in a future study. Whether the trade favors the blade or the kicker must be decided with the ring, not within the line.

4.5 Sensitivity to the error levels

The 0.5 mrad rms quadrupole roll is large for a transfer line, and 0.1 % rms shot energy jitter is at the high end of booster performance. Table 8 repeats the steered evaluation with rolls of 0.1 mrad, 0.2 mrad and 0.5 mrad and energy jitter of 0.05 % and 0.1 % on another 200 independently seeded error realizations (analytic loss and clearance, no particle tracking). Changing the roll level changes the predicted loss by less than 0.001 percentage points, i.e. not at all at the precision reported in Table 8: quadrupole rolls couple a small part of the horizontal motion into the vertical plane, whose restrictions keep several millimeters of clearance, and leave the horizontal blade loss unchanged. Halving the energy jitter lowers the matched-reference loss from 1.34 % to 1.32 %, because its 81 µm exit contribution is small compared with the betatron jitter. The ranking of the settings is the same at every level. On this population the loss-based setting is 5.6 times below the matched reference, against 3.3 on the tracked test population: the size of the gain depends on the sampled error realizations, as expected for a heavy-tailed loss, while its sign does not.

Table 8: Sensitivity of the steered matched line (Matched denotes the matched reference) to the roll and energy-jitter levels: mean predicted loss and mean minimum five-sigma clearance on 200 error realizations with three shots each (analytic, not tracked).

Roll	$\sigma_{\delta,j}$	Predicted loss [%]			$\bar{c}_{\min}$ [mm]	
[mrad]	[%]	Matched	Low beta	Loss-based	Matched	Loss-based
0.1	0.05	1.32	1.20	0.23	0.519	0.504
0.1	0.10	1.34	1.24	0.24	0.517	0.502
0.2	0.05	1.32	1.20	0.23	0.519	0.504
0.2	0.10	1.34	1.24	0.24	0.517	0.502
0.5	0.05	1.32	1.20	0.23	0.519	0.504
0.5	0.10	1.34	1.24	0.24	0.517	0.502

5 Discussion and conclusions

Optimization of the baseline BTS line restores the prescribed exit lattice functions, reduces the large vertical beta peak, and changes the nominal blade clearance from negative to positive. These changes establish the benefit of matching and envelope control relative to the baseline configuration. The subsequent matched-only comparisons address a different question: how much additional improvement is possible once the endpoint optics are fixed.

In a BTS line matched at both ends, the beam size at the ends is fixed and so is the nominal clearance of every restriction outside the quadrupole section. With the blade geometry of the NKM injection study, the injection septum is that restriction, and it limits the line by an order of magnitude more than any interior aperture. Lowering the interior peak beta, the usual design

heuristic, therefore changes only clearances of several millimeters and, on fresh steered error realizations, leaves the loss unchanged.

The quadrupoles act on the loss through the trajectory. Static errors are removed by field trims of the existing dipoles, without adding any element to the line; the two trims nearest the exit steer both position and angle at the septum. What remains is booster shot jitter, and for matched ends its arrival at the exit depends only on the horizontal phase advance. The line chooses whether the jitter appears as position at the blade or as angle at the injection kicker, but it cannot remove it. For the configured exit, matched solutions reach only a 32° interval of phase advance, and the best setting in it reduces the blade loss 3.3-fold at 13 % more angle jitter. A dispersion-free exit widens the interval and lowers the predicted loss much further, again at the cost of angle jitter. The choice between blade loss and injection-angle jitter belongs to the injection design as a whole: for the NKM, whose 80 % angular acceptance is 0.44 mrad, exit-angle jitter of 0.33 mrad–0.39 mrad is of the same order, so the trade must be evaluated with ring capture, and the booster's angle stability bounds what either choice can achieve.

Two methodological points carry over. The mean minimum five-sigma clearance does not rank settings by loss: it rewards static offsets away from a one-sided restriction and is blind to the jitter tail. A Gaussian loss probability from the propagated centroid and covariance agrees with tracked losses within 1 %, is smooth, and can be optimized directly. The phase-advance view also gives a quick test for any matched transfer line: compute μ_x, read off where the booster jitter lands, and compare with the restriction and the kicker acceptance.

The loss is heavy-tailed over error realizations and its population mean varies between independently seeded populations, while the ranking of settings does not. The dispersion-free results are analytic. This paper neither certifies real transmission nor derives construction tolerances, ring capture, or kicker transparency.

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